

Recap: Causal Bellman Lower Bound

- ▶ Unobserved confounding $\Rightarrow P(s' \mid s, do(x))$ not identifiable
- ▶ Per-step Manski bound (with local neighborhood $\mathcal{N}(s, x)$):

$$\mathbb{E}[V(S')] \geq P(x \mid s) \mathbb{E}_{P_{\text{obs}}}[V(S')] + (1 - P(x \mid s)) \min_{s'' \in \mathcal{N}(s, x)} V(s'')$$

- ▶ Causal Bellman lower-bound equation:

$$\begin{aligned} \underline{d}^\pi(s_g \mid s, x) = & (1 - \gamma) \mathbb{I}[s = s_g] \\ & + \gamma P(x \mid s) \sum_{s'} P_{\text{obs}}(s' \mid s, x) \underline{V}_d(s') \\ & + \gamma (1 - P(x \mid s)) \min_{s'' \in \mathcal{N}(s, x)} \underline{V}_d(s'') \end{aligned}$$

Reweighted NCE for Sampling Design

Setup: $P(s' \mid s, do(x))$ interventional (not identifiable), $P_{\text{obs}}(s' \mid s, x)$ observational, $P(x \mid s)$ propensity.

Standard contrastive RL (Eysenbach et al., 2022):

$$p^+(s, x, s_f) = \mu(s, x) \cdot d^\pi(s_f \mid s, x)$$

$$p^-(s, x, s_f) = \mu(s, x) \cdot p(s_f)$$

Bayes-optimal critic: $f^* = \log \frac{d^\pi(s_f \mid s, x)}{p(s_f)}$

Our redesign. Replace d^π with $\underline{d}^\pi$:

$$p^+(s, x, s_f) = \mu(s, x) \cdot \underline{d}^\pi(s_f \mid s, x)$$

$$p^-(s, x, s_f) = \mu(s, x) \cdot p(s_f)$$

Bayes-optimal critic: $f^* = \log \frac{\underline{d}^\pi(s_f \mid s, x)}{p(s_f)}$

What needs to be shown:

1. $\underline{d}^\pi$ well-defined (unique fixed point)
2. $\underline{d}^\pi \leq d^\pi$ pointwise
3. Executable sampler for p^+

Theoretical Results

Causal Bellman operator $\underline{T}$ with local neighborhood $\mathcal{N}(s, x)^*$:

$$\begin{aligned}(\underline{T}f)(s, x) = & (1 - \gamma)\mathbb{I}[s = s_g] \\ & + \gamma P(x \mid s) \mathbb{E}_{s' \sim P_{\text{obs}}(\cdot \mid s, x)}[V_f(s')] \\ & + \gamma(1 - P(x \mid s)) \min_{s'' \in \mathcal{N}(s, x)} V_f(s'').\end{aligned}$$

Lemma 1 $\underline{T}$ is a γ -contraction $\Rightarrow$ unique fixed point $\underline{d}^\pi$.

Lemma 2 Per-step Manski: $\underline{T}f \leq \mathcal{T}^*f$ pointwise.

Thm 1 $\underline{d}^\pi \leq d^\pi$ everywhere; valid lower bound.

★ **Thm 2** Recursive sampler: $\Pr(S_f = s_g \mid s, x) = \underline{d}^\pi(s_g \mid s, x)$. ★

★ **Thm 3** Under $(p^+ = \mu \underline{d}^\pi, p^- = \mu p)$, $f^* = \log \frac{\underline{d}^\pi(s_f \mid s, x)}{p(s_f)}$.

Lemma 1 re-proved with $\mathcal{N}(s, x)$ replacing the global minimum.

Supporting Theorems (1/2)

Lemma 1 (γ -contraction). The Causal Bellman operator $\underline{T}$ defined by

$$(\underline{T}f)(s, x) = (1-\gamma)\mathbb{I}[s = s_g] + \gamma P(x \mid s) \mathbb{E}_{T_e}[V_f] + \gamma(1-P(x \mid s)) \min_{s'' \in \mathcal{N}(s, x)}$$

satisfies $\|\underline{T}f - \underline{T}g\|_\infty \leq \gamma \|f - g\|_\infty$.

$\Rightarrow$ Unique fixed point $\underline{d}^\pi$ by Banach.

Lemma 2 (Per-step Manski bound). For $f \geq 0$:

$$(\underline{T}f)(s, x) \leq (\mathcal{T}^*f)(s, x)$$

where $\mathcal{T}^*$ is the standard Bellman operator under the interventional transition $T(\cdot \mid s, do(x))$.

Theorem 1 (Validity).

$$\underline{d}^\pi(s_g \mid s, x) \leq d^\pi(s_g \mid s, x) \quad \forall (s, x, s_g).$$

Proof. $\underline{d}^\pi = \underline{T}\underline{d}^\pi \leq \mathcal{T}^*\underline{d}^\pi$. Iterate $\mathcal{T}^*$ (monotone, γ -contraction with fixed point d^π): $\underline{d}^\pi \leq (\mathcal{T}^*)^k \underline{d}^\pi \rightarrow d^\pi$. ■

Supporting Theorems (2/2)

Theorem 2 (Sampling from $\underline{d}^\pi$). Given $\underline{V}$, starting from $(S_0, X_0) = (s, x)$, recursively generate (S_{t+1}, X_{t+1}) from (S_t, X_t) via:

- (i) w.p. $P(X_t \mid S_t)$: draw $S_{t+1} \sim T_e(\cdot \mid S_t, X_t)$;
 - (ii) w.p. $1 - P(X_t \mid S_t)$: set $S_{t+1} = \arg \min_{s'' \in \mathcal{N}(S_t, X_t)} \underline{V}(s'')$;
- followed by $X_{t+1} \sim \pi(\cdot \mid S_{t+1})$. Let $T \sim \text{Geom}(1 - \gamma)$ independently, and define $S_f := S_T$. Then

$$\Pr(S_f = s_g \mid s, x) = \underline{d}^\pi(s_g \mid s, x).$$

Proof. Let $q(s_g \mid s, x) := \Pr(S_f = s_g \mid s, x)$. Decomposing on $T \sim \text{Geom}(1 - \gamma)$: with probability $1 - \gamma$, $T = 0$ and $S_f = s$; with probability γ , $T \geq 1$ and we sample (S_1, X_1) via (i)(ii) then recurse. This gives $q = \underline{T}q$. Lemma 1 $\Rightarrow$ unique fixed point, hence $q = \underline{d}^\pi$. ■

Theorem 3 (Bayes-optimal critic)

Theorem 3 (Bayes-optimal critic). Under $p^+ = \mu \cdot \underline{d}^\pi$, $p^- = \mu \cdot p$, the NCE-binary objective

$$\mathcal{L}(f) = \mathbb{E}_{p^+}[\log \sigma(f)] + \mathbb{E}_{p^-}[\log(1 - \sigma(f))]$$

admits Bayes-optimal critic

$$f^*(s, x, s_f) = \log \frac{d^\pi(s_f \mid s, x)}{p(s_f)}.$$

Proof. Pointwise stationarity:

$$\sigma(-f^*)p^+ - \sigma(f^*)p^- = 0 \Rightarrow f^* = \log(p^+/p^-).$$



Experiments: MiniGrid WindyCorridor

Setup. Confounded offline data; forced- U evaluation; 50 episodes per regime; seed 0.

Method	$\uparrow$ success $_{u=0}$	$\uparrow$ success $_{u=1}$	$\uparrow$ mean	$\uparrow$ worst-case	$\downarrow$ gap
Confounded baseline	1.00	0.44	0.72	0.44	0.56
Causal Contrastive	0.98	0.56	0.77	0.56	0.42
Oracle upper bound	1.00	0.92	0.96	0.92	0.08

Key observations.

- Worst-case success: 0.44 $\rightarrow$ 0.56

Open Questions

- ▶ **Q1: Theoretical gaps beyond current proofs?**
- ▶ **Q2: Better sampling for p^+ / p^- ?**
- ▶ **Q3: Theory vs. implementation gap?**